

Clarity of Responsibility and Macroeconomic Accountability in Parliamentary Debate

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Abstract

Macroeconomic accountability is usually studied at the moment of electoral reward and punishment. Less attention has been paid to the earlier stage in which parties argue in public over what macroeconomic outcomes mean before votes are cast. This article examines how governing and opposition parties communicate once they enter explicit macroeconomic debate, and when that government-opposition divide becomes sharper. The argument is that governments and oppositions face the same macroeconomic signal from different positions of responsibility, and that this asymmetry should be most visible when governing responsibility is concentrated and easy to attribute. To study this process, a multilingual text-classification pipeline is applied to parliamentary speeches from Austria, the Czech Republic, Germany, Denmark, Spain, Croatia, and Hungary. The analysis distinguishes between positional and valence communication and then disaggregates valence into credit versus blame and internalized versus externalized responsibility attribution. The analysis shows a consistent pattern. Opposition parties rely more heavily on valence overall, whereas governing parties, within the subset of speeches using non-trivial valence, tilt more toward credit and internalized responsibility. Greater clarity of responsibility matters most for overall valence use. The article contributes theoretically by showing how accountability is contested before the vote and empirically by tracing that process in a comparative speech-level design restricted to explicit macroeconomic performance debate.

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Introduction

Macroeconomic accountability is usually studied at the moment of electoral reward and punishment (Duch and Stevenson, 2008, Hellwig, 2010, Lewis-Beck and Stegmaier, 2007). Yet accountability is also fought over before election day. When inflation rises, growth slows, or unemployment worsens, parties argue over what those developments mean, who should own them, and who deserves credit or blame. Parliament is a particularly useful arena in which to study that conflict. Governments and oppositions confront one another there repeatedly, in public, and under visible responsibility constraints (Helms, 2008, Garritzmann, 2017, Proksch and Slapin, 2015, Bäck et al., 2021). The broader government-opposition divide is visible not only in roll-call behaviour, but also in interruptions, attacks, and other forms of discursive conflict (Hix and Noury, 2016, Diener, 2025, Koch, 2025, Poljak, 2023, Poljak and Walter, 2024).

Three literatures speak to this problem, but rarely in the same design. The economic voting literature explains when voters hold incumbents responsible for national performance (Duch and Stevenson, 2008, Hellwig, 2010, Powell, 2000). Research on party communication and valence politics explains how competence, performance, and issue ownership become politically contested (Stokes, 1963, 1992, Clark, 2009, Petrocik, 1996, Green and Jennings, 2017). Research on parliamentary politics and government-opposition competition shows that legislatures are structured by enduring office-status roles that shape conflict, scrutiny, and debate (Strøm, 1990, Strøm et al., 2003, Helms, 2008, Garritzmann, 2017, Hix and Noury, 2016, Proksch and Slapin, 2015, Bäck et al., 2021). What is still missing is a direct account of how governments and oppositions communicate about macroeconomic performance inside parliament, and how that communication changes across accountability environments.

This paper asks two questions. First, once parties are in explicit macroeconomic debate, how does office status shape the balance between positional and valence communication? Second, when do those government-opposition differences become sharper? The paper does not study whether parties choose to raise the economy in parliament in the first place. It studies how they speak once macroeconomic performance is explicitly on the floor. The argument is that governments and oppositions confront the same macroeconomic signal from different positions in the chain of responsibility (Powell and Whitten, 1993, Hobolt et al., 2013). That asymmetry should structure both how much parties rely on valence communication and how they allocate valence once they use it. The paper’s main contextual condition is clarity of responsibility: when it is easier to identify who governs and who can plausibly be held responsible, incumbents become easier to target and have less room to diffuse responsibility.

The analysis draws on parliamentary speeches from Austria, the Czech Republic, Germany, Denmark, Spain, Croatia, and Hungary. It combines a multilingual text-

classification pipeline with speech-level models that distinguish between positional and valence communication and then disaggregate valence into credit versus blame and internalized versus externalized responsibility attribution. Theoretically, the paper shows how economic accountability is contested before the vote. Empirically, it studies that process in a comparative speech-level design restricted to explicit macroeconomic performance debate rather than general economic discussion. Macroeconomic accountability is publicly constructed in parliament before elections, and the government-opposition divide becomes sharper when responsibility is clearer.

Economic Voting, Parliamentary Debate, and Clarity of Responsibility

The starting point is the economic voting literature. A large body of work shows that voters are more likely to reward incumbents when the economy goes well and to punish them when it goes badly (Duch and Stevenson, 2008, Hellwig, 2010, Lewis-Beck and Stegmaier, 2007, Powell, 2000). But that literature mostly observes the end point of the accountability process: electoral judgment. It is less informative about how accountability is built in public before citizens cast a vote. Economic outcomes do not arrive with a political label attached. Parties must argue over what those outcomes mean, whether they reflect competence or failure, and who should own them.

Parliamentary debate is a good setting in which to study that process because it is a public arena where governments defend their record, opposition parties challenge that record, and both sides try to impose an interpretation of responsibility (Helms, 2008, Garritzmann, 2017, Proksch and Slapin, 2015, Bäck et al., 2021). This is especially true in coalition settings, where parliamentary institutions help structure how governing actors defend and justify the coalition record under shared responsibility constraints (Martin and Vanberg, 2011). Parliamentary debate is also relevant beyond the chamber itself. Parliamentary interventions often feed mediated political communication, and parliamentary activity is positively associated with politicians' media visibility (Yildirim et al., 2023). This makes parliament useful not only as a site of intra-elite confrontation, but also as a setting in which public accountability claims are articulated and amplified. This matters especially for the economy because governments cannot easily avoid the topic. Even when the economy is going badly, incumbents still have to speak about inflation, unemployment, growth, taxes, deficits, and budgets. Opposition parties face the same agenda, but from a different strategic position. They can turn the same macroeconomic developments into attacks, warnings, or prospective competence claims of their own. Recent work on parliamentary confrontation shows the same government-opposition logic in interruptions, attacks, forms of address, and other non-substantive clashes (Ilie, 2010,

Diener, 2025, Koch, 2025, Poljak, 2023, Poljak and Walter, 2024).

The distinction between positional and valence communication helps organize this debate. Parties do not compete only over policy alternatives. They also compete over broadly shared goals such as competence, effective performance, responsibility, and credibility (Stokes, 1963, 1992, Clark, 2009). In macroeconomic debate, parties can communicate positionally by arguing over taxes, spending, labour-market reform, or policy trade-offs. They can also communicate through valence by treating outcomes as evidence that someone is competent, irresponsible, honest, ineffective, or unable to govern. In this paper, valence therefore includes both character-based claims and policy- or performance-based claims about who can manage the economy well. This broader treatment is consistent with work in party competition, campaign communication, and parliamentary speech that treats valence as encompassing leadership traits together with performance and issue-management appeals (Clark, 2009, Adams et al., 2005, Bjarnøe et al., 2023, Bäck et al., 2021). Character-based statements emphasize leaders' or parties' personal valence traits without linking them to a specific policy, whereas policy-valence statements connect those same valence concepts to concrete domains such as inflation, growth, or unemployment (Abney et al., 2013). This operationalization is consistent with related valence datasets and improves conceptual clarity across parties, countries, and time periods (Debus et al., 2018). When competence or responsibility language turns macroeconomic discussion into an accountability claim, the unit is analytically evaluative rather than purely positional.

Valence also has an internal structure. Once parties move from policy disagreement to evaluative claims, they still have to decide what kind of claim to make. They can claim credit or assign blame. They can attach responsibility to their own side or push it onto opponents, inherited conditions, institutions, or external constraints (Weaver, 1986, Hood, 2011, Tilley and Hobolt, 2011, Hellwig, 2001). These choices are central to accountability. They tell citizens not only whether the economy is being described positively or negatively, but also who is said to own the outcome. The paper therefore speaks both to valence theory and to issue ownership research (Petrocik, 1996, Green, 2007, Belanger and Meguid, 2008, Wagner and Meyer, 2014, Green and Jennings, 2012, 2017).

The baseline expectation follows from office status. Governments and opposition parties face the same economic signal, but they do so from different responsibility positions (Powell and Whitten, 1993, Duch and Stevenson, 2008, Hobolt et al., 2013). Incumbents are more exposed to accountability for national performance. That gives them stronger reasons to defend their record and to connect favorable outcomes to their own competence. But incumbency also constrains them. Governing parties cannot live on competence claims alone; they must justify policy choices, defend trade-offs, and answer for outcomes they cannot fully disown. Opposition parties face a looser constraint.

They can more easily turn the same macroeconomic developments into blame, criticism, or prospective competence claims against the government (Nai and Maier, 2024). The asymmetry should therefore appear both in overall valence use and in the way valence is allocated once it is used.

H1. Within macroeconomic parliamentary debate, opposition parties should rely more heavily on valence overall than governing parties.

The main contextual argument concerns clarity of responsibility. The core insight of the clarity literature is that accountability becomes easier when governing responsibility is more concentrated and more visible (Powell and Whitten, 1993, Duch and Stevenson, 2008, Hobolt et al., 2013). Following Hobolt et al. (2013), it is useful to distinguish between institutional clarity and government clarity. This paper focuses on the second dimension. The relevant question here is whether there is an identifiable and cohesive incumbent to whom macroeconomic outcomes can plausibly be attached. That should matter not only at election time, but also inside parliamentary debate.

Clarity of responsibility should sharpen this divide through two linked mechanisms. For opposition parties, clearer responsibility makes attack more efficient because macroeconomic outcomes can be attached to a more legible incumbent target (Rudolph, 2003). For governing parties, the same concentration of responsibility tightens ownership constraints: as accountability becomes harder to diffuse, incumbents have less room to shift responsibility away and greater need to defend concrete records, choices, and trade-offs. Clearer responsibility should therefore sharpen the government-opposition divide in macroeconomic communication. The strongest expectation concerns overall valence use. If clearer responsibility also changes how parties compose valence once they use it, that should appear secondarily in credit claims and responsibility attribution.

H2. As clarity of responsibility increases, the government-opposition gap in overall valence use should widen.

Office status should also shape the internal composition of valence. Once governing parties adopt an evaluative frame, the most credible positive claims available to them are typically claims of owned performance: they can present favorable outcomes as evidence of their competence and attach responsibility to their own side. Opposition parties face the mirror incentive. Because they are not responsible for the incumbent record, their most available evaluative claims are negative and externally directed, casting governments as the locus of failure. Governments can of course attempt externalization under unfavorable conditions, but conditional on using valence at all, incumbency should make credit and internalized attribution more defensible rhetorical options for them than for opposition parties (Vis, 2016).

H3. Conditional on using valence, governing parties should allocate more of that valence toward credit and internalized responsibility than opposition parties.

Data & Methods

The Dataset

The empirical analysis draws on parliamentary speech corpora from seven European parliamentary democracies: Austria (AT), the Czech Republic (CZ), Germany (DE), Denmark (DK), Spain (ES), Croatia (HR), and Hungary (HU). The speech texts and metadata come from the ParlLawSpeech corpus (Schwalbach et al., 2025), which provides harmonized records of plenary interventions across countries, including speech identifiers, dates, and party attribution. The broader comparative data-collection logic follows recent work on large-scale legislative debate corpora (Schwalbach and Rauh, 2021). More generally, the paper treats validation as a multilingual design issue rather than as a single classifier score reported at the end of the workflow (Wilkerson and Casas, 2017, Barberá et al., 2021, Lind et al., 2025).

The case selection is designed to balance institutional comparability and contextual variation. All seven cases are parliamentary democracies in which governments are publicly accountable for national economic performance, yet they vary substantially in coalition structure, party-system organization, and economic context. The aim is not to estimate a universal causal effect of context across all parliamentary systems, but to assess whether the same broad accountability logic can be observed across diverse parliamentary settings within a common design.

The full corpus is filtered in two ways before substantive modeling. First, entries with non-informative party placeholders are removed. Second, speeches shorter than 80 words are excluded in order to reduce procedural fragments, interruptions, and other low-information observations. After these filters, the seven-country corpus contains 981,731 speeches. Of these, 75,528 are retained as macroeconomic speeches and form the basis of the final speech-level analysis panel, spanning the period from 1994 to 2023. Coverage varies across countries, with the earliest observations in Hungary and the latest in the Czech Republic and Spain.

Table 1: Country coverage in the speech-level analysis panel

Country	Speeches	First speech	Last speech
AT	10,719	1996-01-31	2019-09-25
CZ	2,729	2013-11-27	2023-10-11
DE	7,134	2009-11-10	2021-09-07
DK	9,670	2007-11-27	2022-10-06
ES	20,654	1996-05-03	2023-07-26
HR	8,603	2003-12-22	2020-05-15
HU	16,019	1994-06-28	2022-03-10
Total	75,528	1994-06-28	2023-10-11

Pipeline overview

Figure 1 summarizes the measurement strategy. The design mirrors the theoretical distinction between the overall prominence of valence and its internal allocation once parties adopt that rhetorical strategy.

In a first step, the analysis is restricted to speeches that explicitly address macroeconomic conditions. In a second step, each speech is segmented into quasi-sentences, which serve as the basic unit for text classification. In a third step, a stage-one classifier assigns each quasi-sentence to one of three categories: positional, valence, or other. In a fourth step, quasi-sentences classified as valence are passed to stage-two models that distinguish credit from blame and internalized from externalized responsibility attribution. Finally, the quasi-sentence predictions are aggregated back to the speech level and merged with contextual variables.

This hierarchy matters substantively. The paper does not treat macroeconomic speech as a single rhetorical mode. It first asks whether a speech relies on valence at all, and then asks what kind of valence is used once that strategy appears.

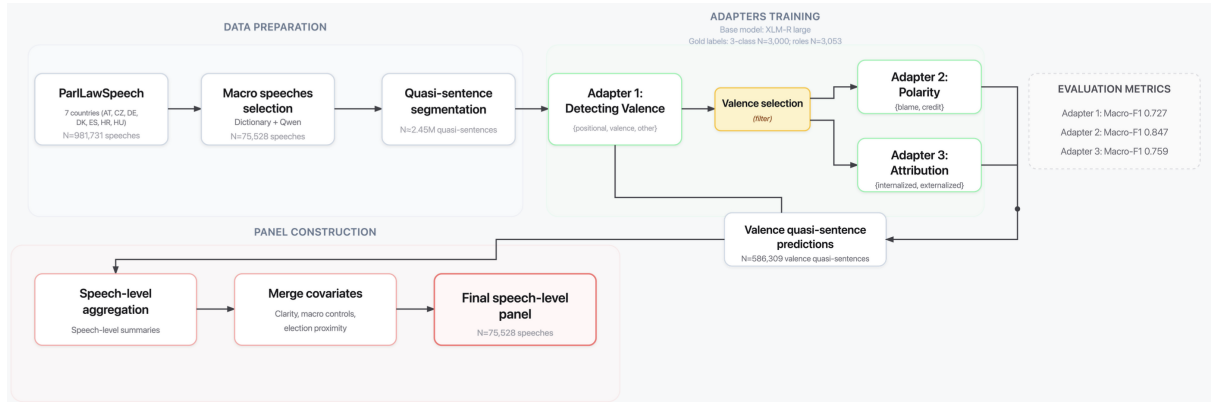


Figure 1: End-to-end pipeline from parliamentary speeches to the final speech-level analysis panel.

Identifying macroeconomic speeches

The analysis is restricted to speeches that explicitly address aggregate macroeconomic performance. These speeches are identified using a transparent rule-based gate built from multilingual dictionaries. The gate searches full speech text for references to three macroeconomic domains: GDP, inflation, and unemployment, but the dictionaries extend beyond exact indicator labels to related formulations such as growth, recession, rising prices, cost of living, and labour-market conditions.¹ A speech is classified as

¹The multilingual term lists were assembled by the author with input from language experts for each corpus language and then implemented as country-aware, case-insensitive regular expressions with boundary-safe matching. The full lists are reported in Appendix C.

macroeconomic if it contains at least one valid match in any of these domains. Each flagged speech is also tagged by macroeconomic category.

The topical gate serves both a substantive and a methodological purpose. Substantively, it ties the analysis to aggregate macroeconomic performance rather than to economic policy discussion in general. Methodologically, it preserves interpretability and transparency, because every inclusion decision can be traced to explicit lexical criteria. The gate therefore defines the scope of the estimand. It identifies the subset of parliamentary speech in which macroeconomic performance is explicitly invoked and therefore publicly available as an accountability object; it is not intended to recover the full universe of economy-related rhetoric. Its three domains, growth, inflation, and unemployment, operationalize explicit references to aggregate macroeconomic performance rather than to general economic policy. They are selected because they are the canonical macroeconomic outcomes around which economic accountability is conventionally organized in the economic-voting literature. The design therefore prioritizes scope validity and precision over exhaustive recall.

This distinction matters for interpretation. Economic conditions may shape both whether parties enter macroeconomic debate and how they speak once they do. The present analysis addresses only the second margin. The design is therefore conditional by construction, and all subsequent claims concern rhetorical strategy in parliamentary macroeconomic speech rather than agenda entry into that debate. Speeches about fiscal policy, welfare, energy, or other domains that never invoke explicit macroeconomic performance are outside the paper’s estimand by design.

As additional validation, Appendix C reports both a manual quality-control check and a comparison with a Qwen-based topical classifier (Bai et al., 2023). Those checks are used to assess scope and transparency, not to redefine the gate itself. The rule-based approach remains the preferred filter because it is transparent, reproducible, and closely matched to the paper’s estimand. A model-based gate would move additional measurement uncertainty into sample definition itself, whereas the present design keeps sample inclusion inspectable and uses machine learning only for the downstream rhetorical classification task.

Quasi-sentence segmentation. After macroeconomic speeches are selected, full-text records are reconstructed and segmented into quasi-sentences using deterministic rules. Quasi-sentence segmentation reduces category mixing within sentences and aligns the unit of classification with the distinctions developed in the theory section. This choice follows the quasi-sentence tradition in political text analysis, where texts are decomposed into minimal units expressing a distinct political idea (Klingemann et al., 2006, Mikhaylov et al., 2012). A quasi-sentence is the smallest textual unit to which a class is assigned.

Segmentation proceeds by sentence-final punctuation and hard line breaks. To avoid

over-fragmentation, colons, semicolons, and dashes are left intact by default. A minimum-token rule then iteratively merges short fragments with neighboring text until no fragment shorter than 12 tokens remains or only one fragment is left. The procedure is deterministic and conservative by design. It yields 2,448,800 quasi-sentences mapped to 75,528 macroeconomic speeches.

Speech labeling

The annotation strategy follows a two-stage design. In Stage 1, a stratified gold sample of 3,000 quasi-sentences is manually labeled into three mutually exclusive classes: positional, valence, and other. This stage identifies whether macroeconomic debate is framed primarily in terms of policy alternatives, competence and responsibility, or neither. The stage-one coding rules are reported in Appendix A, which also provides brief illustrative examples. They capture two complementary dimensions of valence. First, they include character-based valence, such as competence, responsibility, credibility, seriousness, or integrity. Second, they include performance- or issue-based valence, that is, claims about a party’s or government’s ability to manage a valued policy domain effectively. In the macroeconomic context, this includes statements presenting political actors as capable or incapable of delivering growth, employment, price stability, or sound economic management. By contrast, positional units focus on policy instruments, legislative measures, distributive priorities, or alternative economic strategies. The residual “other” category includes procedural, technical, and ceremonial material without substantive framing. A statement such as “we need a different tax policy to bring inflation down” is positional, whereas “this government has shown it cannot control inflation” is valence. Where policy discussion and competence claims appear in the same quasi-sentence, the coding rule gives precedence to valence because the key feature of the unit is evaluative accountability rather than policy positioning.

In Stage 2, a dedicated valence-roles gold set is constructed on valence-only quasi-sentences ($N = 3,053$), stratified across countries. The stage-two codebook defines two role dimensions. The first is polarity: blame versus credit. The second is responsibility attribution: internalized versus externalized. The target is the actor or locus to which causal responsibility is assigned, rather than any actor mentioned in passing. Internalized attribution refers to own-side ownership of responsibility: performance or failure is attributed to the speaker’s own party, coalition, ministers, or government where that ownership is explicit. Externalized attribution refers to the displacement of responsibility outside the speaker’s own side, most often onto other domestic political actors and,

less frequently, onto institutions or extra-domestic constraints.² For annotation support, quasi-sentences are machine-translated into English using Qwen before coding (Bai et al., 2023). Translation is used only to support human annotation; the production classifiers operate on the original multilingual text (de Vries et al., 2018).

Adapter training and classification

All text models are multilingual XLM-RoBERTa-large classifiers (Conneau et al., 2020) fine-tuned with LoRA adapters (Poth et al., 2023). One main three-class adapter is trained for the positional/valence/other classification task, followed by two role-specific adapters on valence-only quasi-sentences: a polarity adapter distinguishing blame from credit and an attribution adapter distinguishing internalized from externalized responsibility. Train, development, and test splits are grouped at the speech level in order to prevent leakage across partitions. Recent work on multilingual computational text analysis stresses that these validation choices should be made explicit across the workflow rather than being left implicit behind a single final metric (Licht, 2023, Wang, 2023, Lind et al., 2025). Detailed held-out performance metrics are reported in Appendix B.

Held-out performance is strongest for polarity, satisfactory for attribution, and more modest for the main three-class task. Appendix B reports the full metrics. For the stage-one classifier, held-out macro-F1 under the default argmax rule is 0.727. Valence precision is 0.613 and valence recall is 0.691. The paper then applies a thresholded decision rule for the valence class: a quasi-sentence is coded as valence only if its predicted valence probability exceeds 0.56; otherwise it is assigned to the most likely non-valence class. Threshold tuning of this kind is standard when the target class is substantively important and the costs of overprediction and underprediction are asymmetric (Lipton et al., 2014, Berger and Guda, 2020). In this case, the rule preserves macro-F1 while increasing valence precision to 0.648 and aligning the predicted valence share more closely with the held-out gold distribution. The stage-one classification problem is demanding because valence often overlaps with substantive policy discussion, especially in macroeconomic speech where competence claims are embedded in debate over concrete measures, outcomes, and trade-offs (Stokes, 1992, Clark, 2009). Stage one is therefore the main measurement bottleneck, and the analysis is designed accordingly at the speech level. The appendix reports both the threshold audit and the resulting speech-level outcome distribution.

Stage one is also the conceptually hardest task: separating positional from valence language in multilingual macroeconomic debate has no simple off-the-shelf analogue. The paper does not depend on perfect quasi-sentence classification. Its inferential target

²Although the externalized category conceptually includes institutional and extra-domestic attributions, most labeled externalized cases refer to other domestic political actors. More distal attributions, such as supranational constraints or impersonal shocks, are comparatively rare, so the category is not split further.

is speech-level rhetorical composition, and aggregation to the speech level reduces the influence of individual quasi-sentence errors. The substantive claims therefore rest on the stability of the speech-level pattern across the appendix checks rather than on classifier performance alone.

Dependent variables: speech-level measures

The empirical analysis focuses on two related strategic margins of macroeconomic communication. The first concerns how strongly a speech relies on valence rather than on positional or residual content. The second concerns how parties compose valence once that rhetorical mode is used. This distinction follows directly from the theory section.

The first outcome is the overall valence share of the speech. Let Q_s denote the total number of quasi-sentences in speech s , and V_s the number of quasi-sentences classified as valence. The overall valence share is

$$Y_s = \frac{V_s}{Q_s}.$$

Higher values therefore mean that a larger share of the speech is devoted to evaluative claims about competence, performance, or responsibility. Lower values do not mean that parties are speaking less; they mean that more of the speech is allocated to positional or residual forms of communication.

The remaining outcomes capture how valence is allocated when it is present. Two dimensions are central: polarity, that is, whether valence is used to claim credit or to assign blame, and responsibility attribution, that is, whether performance is owned by the speaker’s own side or displaced onto actors and forces outside it. These role-specific outcomes are defined within the valence subset of each speech.

Let B_s , C_s , I_s , and E_s denote the numbers of blame, credit, internalized, and externalized quasi-sentences in speech s , respectively. The corresponding conditional shares are defined as

$$\begin{aligned} B_s^* &= \frac{B_s}{V_s}, & C_s^* &= \frac{C_s}{V_s}, \\ I_s^* &= \frac{I_s}{V_s}, & E_s^* &= \frac{E_s}{V_s}. \end{aligned}$$

In the main text, the emphasis is placed on three outcomes: the overall valence share Y_s , the credit share C_s^* , and the internalized share I_s^* . This keeps the empirical presentation aligned with the core theoretical distinction between the amount of valence and its internal allocation. Credit and blame are complementary within the polarity dimension, and internalized and externalized attribution are complementary within the attribution dimension. The mirror outcomes are therefore retained for appendix robustness checks rather than treated as separate main-text outcomes.

These outcomes capture different strategic margins. Whereas Y_s indicates how much a speech relies on valence at all, the conditional measures indicate how that valence is allocated after the speech adopts a competence-based frame. A speech may contain relatively little valence overall but still allocate that limited valence heavily toward blame or toward externalization; conversely, a strongly valence-oriented speech may divide its evaluative content more evenly across rhetorical roles.

Because the composition measures are meaningful only when a speech contains a non-trivial amount of valence content, the corresponding models are estimated on speeches with at least three valence quasi-sentences. This threshold reduces instability in ratios based on very small denominators while preserving the speech-level interpretation of composition. It also means that the composition models are explicitly conditional: they describe how parties allocate valence once a non-trivial amount of valence is present, not whether parties select into using valence in the first place. Appendix C characterizes how speeches above and below the composition threshold differ descriptively.

Independent variables: office status, clarity, and controls

The empirical strategy is organized around one central contextual mechanism: clarity of responsibility. The paper’s main question is whether governments and oppositions communicate differently about macroeconomic performance, and whether that divide becomes more visible when responsibility is easier to assign.

Government status is constructed from cabinet-party membership information provided by ParlGov (Döring and Manow, 2024). Speech dates are matched to the relevant cabinet period through a country-party crosswalk, and parties are coded as governing when they are members of the cabinet in office on the date of the speech. Observations for which party mapping cannot be established reliably are treated as missing rather than being mechanically coded as opposition, and these cases are excluded from complete-case model estimation. This coding strategy reduces the risk of misclassifying cabinet responsibility in the pooled analysis.

The political moderator is a clarity-of-responsibility index. More precisely, it operationalizes the government-clarity component emphasized by Hobolt et al. (2013), not the broader institutional-clarity dimension. The index summarizes three features of the governing arrangement: the number of parties in government, the seat share of the prime minister’s party, and the concentration of seats within the governing coalition. Each component maps directly onto the intended mechanism. Responsibility is harder to assign when more parties share office, easier to assign when one prime-ministerial party dominates the cabinet, and easier to assign when governing seats are concentrated rather than dispersed across coalition partners. The operational definition is an equally weighted

average of the pooled-sample standardized components,

$$\text{Clarity}_{it} = \frac{z(-\text{GovParties}_{it}) + z(\text{PM Seat Share}_{it}) + z(\text{Coalition HHI}_{it})}{3}.$$

Standardizing each component on the pooled speech-level sample puts the three ingredients on a common metric before aggregation and makes a one-unit shift in the final index interpretable as a one-standard-deviation shift in the observed speech-level distribution of government clarity. Higher values therefore indicate that responsibility for government performance is more concentrated and easier to attribute. Conceptually, the index captures how easy it is, at the moment a speech is delivered, to identify who governs and who can plausibly be held responsible for macroeconomic outcomes.³

The models also include three lagged country-quarter macroeconomic controls: quarter-on-quarter GDP growth, quarter-on-quarter inflation, and quarter-on-quarter change in unemployment. GDP information comes from the quarterly national accounts aggregates dataset (Eurostat, 2026b), inflation from the monthly Harmonised Index of Consumer Prices dataset (Eurostat, 2026a), and unemployment from the monthly unemployment-rate dataset (Eurostat, 2026c). Speeches observed in quarter t are linked to the macroeconomic controls measured in quarter $t - 1$. The quarter-on-quarter versions are used in the baseline specification because they align more closely with the short-run information environment surrounding parliamentary speech; Appendix D shows that the main clarity result is substantively unchanged when the same controls are introduced in year-on-year form instead.

Finally, the core specification includes election proximity as an additional political control. This variable captures the location of the speech within the parliamentary inter-election cycle and is derived from ParlGov election dates. It is included additively to account for the possibility that parties adjust rhetorical style as elections approach, independently of government status and contextual moderation.

Empirical Strategy

The empirical strategy examines whether the structure of macroeconomic speech differs systematically between governing and opposition parties, and whether that office-status asymmetry varies with clarity of responsibility. The unit of analysis is the speech-level observation. The principal outcome is the overall valence share of the speech, and parallel specifications examine how valence is allocated when that mode is present.

The main specification centers the analysis on office status, clarity of responsibility,

³Appendix C documents the construction in detail, reports the empirical alignment of the three components, and plots the country-quarter trajectories used in the baseline design.

and their interaction, while entering the macroeconomic variables additively as controls:

$$Y_{icpq} = \alpha + \beta_1 \text{InGov}_{icpq} + \beta_2 \text{Clarity}_{icpq} + \beta_3 (\text{InGov}_{icpq} \times \text{Clarity}_{icpq}) \\ + \mathbf{X}'_{c,q-1} \boldsymbol{\delta} + \gamma \text{ElectionProximity}_{icpq} + \mu_p + \lambda_t + \varepsilon_{icpq}. \quad (1)$$

Here, Y_{icpq} denotes the speech-level outcome for speech i , in country c , party p , and quarter q . Depending on the specification, the outcome is either the overall valence share of the speech or one of the main composition outcomes introduced above, namely the credit share within valence or the internalized share within valence. InGov_{icpq} indicates whether the speaker's party is in government. Clarity_{icpq} denotes the clarity-of-responsibility index. $\mathbf{X}_{c,q-1}$ contains the lagged macroeconomic controls for GDP growth, inflation, and unemployment change. μ_p are party fixed effects and λ_t are year fixed effects.

The main quantity of interest is the interaction between government status and clarity of responsibility. It captures whether the government-opposition contrast becomes larger or smaller when responsibility is easier to assign. Party fixed effects absorb stable differences across parties, including persistent ideological and organizational features. Year fixed effects absorb period shocks common across the sample, including broad temporal changes in the macroeconomic and political environment.

The main specifications are linear fixed-effects models with standard errors clustered by country. Linear models are used because they allow the interaction structure to be interpreted transparently across multiple speech-level outcomes while preserving a common specification across the main and composition models. Since the outcomes are bounded, Appendix D reports weighted quasi-binomial fixed-effects models as a check on functional-form dependence. The baseline model is estimated on the three main outcomes emphasized in the theory section: overall valence share, credit share within valence, and internalized attribution within valence. The mirror outcomes are reserved for Appendix D rather than treated as separate main-text outcomes.⁴

The pooled models are designed to recover comparative patterns rather than fully identified causal effects. That caution is especially important because the number of countries is limited and inference with only seven country clusters should be interpreted carefully. The paper therefore relies more on effect sizes, directional stability, and the pattern across robustness checks than on p-values alone. Appendix D adds a country-level few-cluster check in the spirit of Ibragimov and Muller (2016). The point is to assess whether the office-status asymmetry is systematically stronger or weaker under different accountability environments. The clarity interaction is the most informative contextual estimate because it varies within countries over time. Additional checks for measurement,

⁴The macroeconomic variables are entered as additive controls in the baseline design rather than as interacted contextual moderators.

mirror outcomes, topical-gate validation, alternative estimators, country dependence, and exploratory heterogeneity are reported in the appendix.

Results

Table 2 reports the main speech-level fixed-effects models. The valence-share model uses the full macroeconomic-speech panel. The credit and internalized models use the conditional sample of speeches with at least three valence quasi-sentences. All three columns include party and year fixed effects, the same lagged macroeconomic controls, and standard errors clustered by country. Figure 2 summarizes the baseline office-status contrast and the interaction between government status and clarity of responsibility. Figure 3 shows the implied government-opposition gap across the observed range of clarity.

Table 2: Speech-level fixed-effects models of valence prominence and composition

	(1) Valence share	(2) Credit share	(3) Internalized share
In government	-0.134*** (0.012)	0.197*** (0.031)	0.172*** (0.026)
Clarity of responsibility	0.015* (0.006)	-0.002 (0.012)	-0.005 (0.010)
Gov $\times$ Clarity	-0.047** (0.013)	0.055 (0.040)	0.045 (0.033)
GDP growth (QoQ, lagged)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
Inflation (QoQ, lagged)	0.003 (0.002)	-0.001 (0.004)	-0.002 (0.004)
Unemployment change (QoQ, lagged)	0.002 (0.002)	-0.004 (0.003)	-0.001 (0.002)
Election proximity	0.009*** (0.002)	-0.009*** (0.001)	-0.010*** (0.001)
Unit of analysis	Speech	Speech	Speech
Party FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
SE clustered by country	Yes	Yes	Yes
Observations	67,930	42,926	42,926
Within R^2	0.060	0.088	0.055

Because the outcomes are bounded shares, coefficients should be read as shifts in the expected proportion of a speech allocated to valence or to particular forms of valence. On that scale, moving from opposition to government is associated with roughly thirteen fewer percentage points of valence in the average macroeconomic speech. In the composition models, the same shift is associated with roughly twenty more percentage points of credit and seventeen more points of internalized responsibility within the valence portion of the speech.

The baseline office-status asymmetry is clear. Net of party and year fixed effects, speeches by governing parties contain substantially less valence overall than speeches by opposition parties. This is the paper’s first substantive result. Politically, it fits the logic of asymmetrical responsibility costs. Opposition parties can use macroeconomic debate more freely to attack performance, whereas governing parties must also defend policy choices, justify trade-offs, and answer for outcomes they cannot easily disown. The descriptive distribution is consistent with that contrast: the average valence share is 0.269 in opposition speeches and 0.169 in government speeches.

The composition models move in the expected direction. Among speeches with at least three valence quasi-sentences, governing parties allocate more of that valence to credit and to internalized responsibility. The substantive implication is that governments do not merely use less valence overall; they use a different kind of valence once they adopt an evaluative frame. Opposition parties are more valence-heavy in aggregate, but governments, once they rely on valence, are more likely to claim ownership and attach performance to their own side. These composition differences are meaningful at baseline, but they are secondary to the overall valence result and are estimated with greater uncertainty.

Political clarity is the strongest contextual moderator in the analysis. The interaction between government status and clarity of responsibility is negative in the valence-share model and positive in both composition models, although the latter are estimated with substantially wider uncertainty. Figure 3 shows the same pattern in marginal terms: as responsibility becomes more concentrated and more visible, the government-opposition gap widens most sharply for overall valence use, while the corresponding composition margins move in the expected direction but remain less decisive.

This is the paper’s main contextual result. In clearer responsibility settings, opposition parties become even more valence-heavy relative to governing parties. That pattern is consistent with the argument that clarity makes incumbents easier to target and gives them less room to diffuse responsibility. Substantively, the interaction estimate implies that a one-standard-deviation increase in clarity widens the government-opposition valence gap by about 4.7 percentage points. That is roughly one-third of the baseline office-status gap in valence and just under one-third of the average within-party over-time variation in valence share. Across the observed 10th to 90th percentile of clarity, the predicted government-opposition gap rises from about 8.7 to 18.2 percentage points. In practical terms, that is the difference between a speech in which the government-opposition contrast is noticeable and one in which it becomes a dominant feature of the rhetorical mix. The composition outcomes move in the same direction, but the evidence there is weaker and should be read as secondary rather than co-equal to the valence-share result. Clearer responsibility therefore sharpens the partisan structure of macroeconomic accountability communication above all at the level of overall valence prominence.

That interaction should nonetheless be interpreted with some discipline. Clearer responsibility may also coincide with related features such as stronger coalition cohesion, greater message discipline, or a more straightforward opposition target. The observed pattern is consistent with the accountability argument developed here, but it does not by itself prove that no related mechanism is also at work.

The main ordering remains stable across the principal robustness checks reported in Appendix D and Appendix E.⁵ The negative clarity interaction for valence share is especially stable; the composition interactions keep the expected sign but remain more weakly estimated.

Appendix E adds a set of exploratory scope-condition checks. A party-family breakdown suggests that the clarity interaction is most visible among mainstream governing families, especially social democratic and conservative/christian democratic parties. A populist-party split based on Zulianello’s PopulisTree taxonomy (Zulianello, 2026) points in the same general direction, but the difference between populist and non-populist parties is not large enough to displace the baseline clarity-centered account. The appendix also reports an intra-coalition check among governing speeches only. That split suggests that the main government-opposition result is not simply driven by junior coalition partners, although the within-government pattern is clearly secondary to the baseline office-status divide.

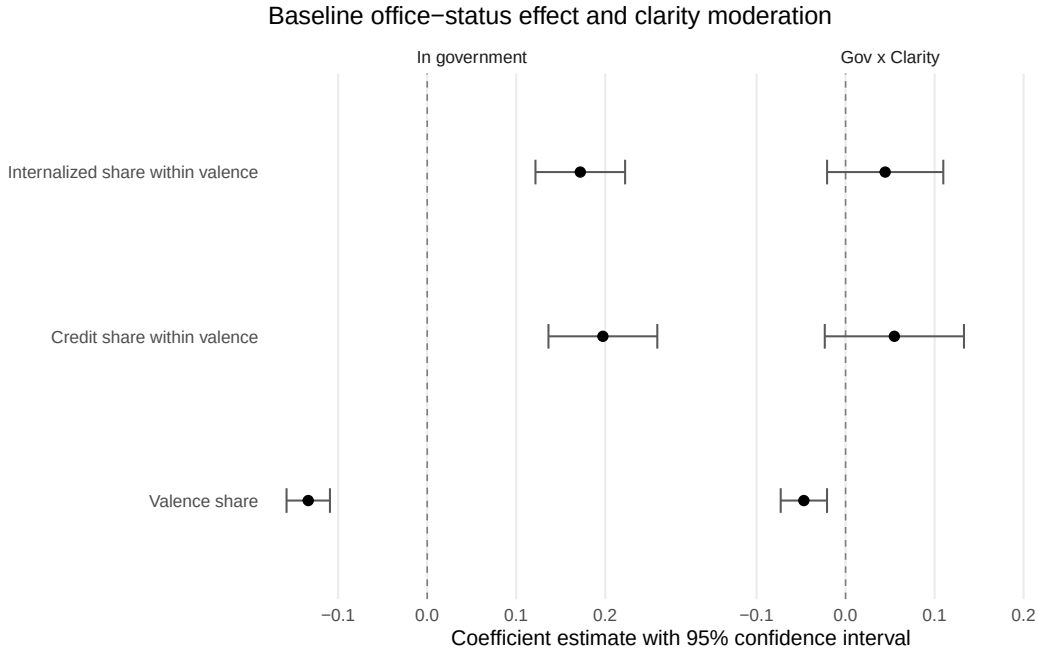


Figure 2: Baseline estimates showing that opposition speeches are more valence-heavy and that the gap widens as clarity increases.

⁵Those checks include same-sample estimation, country-date fixed effects, a country-level few-cluster exercise, leave-one-country-out re-estimation, a within-party switcher restriction, all-speech composition outcomes, and a positional-share mirror model. With only seven country clusters, this appendix pattern matters at least as much as any single conventional significance threshold.

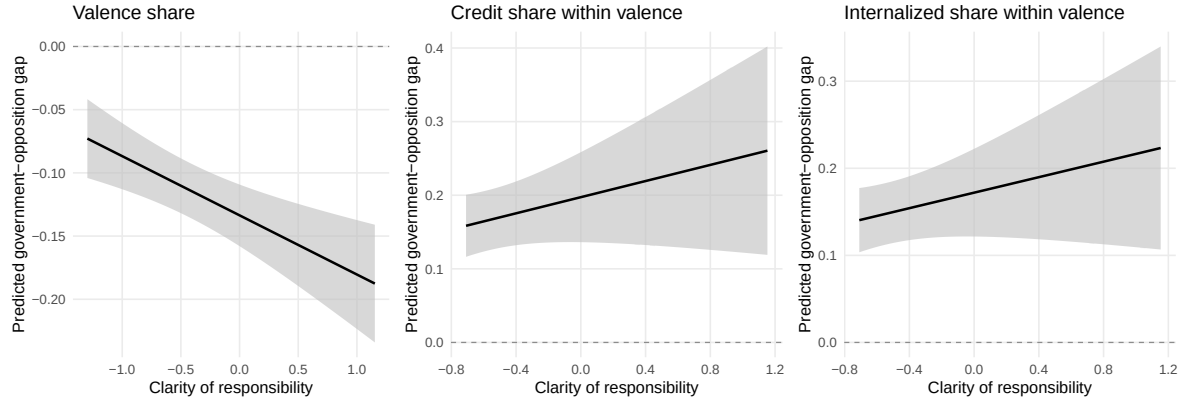


Figure 3: Predicted government-opposition valence gap across clarity of responsibility.

Appendix B documents the measurement pipeline, including the thresholded valence rule. Appendix D and Appendix E then report the main design checks and exploratory heterogeneity splits. Taken together, these checks show that the office-status asymmetry is stable and that the clarity interaction is the paper’s most robust contextual estimate.

Conclusion

This paper studies how governments and oppositions communicate once they explicitly enter macroeconomic debate in parliament. The central claim is that economic accountability is not only registered at election time; it is also constructed in public debate, where parties try to attach outcomes to competence, blame, and ownership.

The evidence shows a consistent pattern. Office status is the baseline divide. Opposition parties use more valence overall, whereas governing parties, once they adopt an evaluative frame, tilt more toward credit and internalized responsibility. The effect of clarity is strongest for overall valence use. When it is easier to identify who governs and who can plausibly be held responsible, the government-opposition divide becomes more visible. That result is strongest for overall valence use and more tentative for the internal composition of valence.

These findings matter for three reasons. For research on economic voting, they identify one pathway through which economic conditions are turned into accountability claims before votes are cast (Duch and Stevenson, 2008, Hellwig, 2010). For research on party communication, they show that the key variation lies not only in whether evaluative language appears, but also in how praise, blame, and responsibility are organized once it does (Stokes, 1992, Petrocik, 1996, Green and Jennings, 2017). For research on parliamentary politics, they show that legislative speech is one of the arenas in which governments and oppositions publicly negotiate responsibility (Proksch and Slapin, 2015, Bäck et al., 2021). More broadly, the paper connects electoral accountability, party communication, and parliamentary competition within a single comparative design.

The paper also has clear limits. The pooled models recover comparative patterns rather than fully identified causal effects. The number of countries is small, and some contextual variation is necessarily correlated with broader country differences. The composition models are intentionally conditional on non-trivial valence use, which fits the theoretical question but does not address prior selection into macroeconomic debate itself. Even so, the central pattern is stable across the main robustness checks and points to the same conclusion: accountability in the economy is shaped not only by outcomes, but also by how clearly responsibility can be attached to those in office and by how parties choose to argue about that responsibility in parliament.

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A Coding Rules and Annotation Design

This appendix documents the coding logic used in the two-stage annotation design. The aim is not to reproduce the full annotation manual, but to make the decision rules behind the speech-level outcomes transparent enough for readers to see how the empirical categories map onto the paper’s theoretical argument.

Stage-one coding rules: positional, valence, and other

The stage-one annotation distinguishes among three mutually exclusive categories.

Positional. Units are coded as positional when they focus on specific policies, laws, programs, fiscal measures, regulatory instruments, or concrete proposals that should be adopted, amended, or rejected. This includes discussion of taxes, welfare, education, energy, budgets, labour-market reforms, and similar policy content. Economic outcomes are coded as positional when they are discussed primarily as expected or observed effects of particular policy instruments. For example, “we should cut payroll taxes to reduce unemployment” is positional.

Valence. Units are coded as valence when they emphasise broadly desirable traits or shared goals, including competence, responsibility, credibility, seriousness, honesty, integrity, unity, or governing capacity. This includes both character-based valence and policy-based valence. In the macroeconomic context, statements presenting political actors as capable or incapable of delivering growth, employment, price stability, or sound economic management are coded as valence. For example, “this government has shown it cannot manage inflation” is valence.

Other. Units are coded as other when they are procedural, technical, or ceremonial and contain no meaningful positional or valence content. This includes agenda-setting remarks, voting instructions, greetings, condolences, and related material.

Decision rules. When economic results are invoked as evidence of competence or failure, the unit is coded as valence. When economic results are discussed as factual or expected consequences of a policy instrument, the unit is coded as positional. If a quasi-sentence contains both policy discussion and a meaningful competence or character claim, valence takes precedence because the paper is interested in whether parties convert substantive macroeconomic debate into evaluative accountability claims.

Stage-two coding rules: polarity and responsibility attribution

Stage 2 is applied only to units already classified as valence.

Polarity. Valence units are coded as credit when they make a positive evaluative claim about political performance or character, including competence, reliability, honesty, leadership, or successful governance. They are coded as blame when they make a negative evaluative claim about performance or character, including incompetence, dishonesty, unreliability, failure, or mismanagement.

Responsibility attribution. Valence units are coded for the main attribution channel of causal responsibility. Attribution is coded as internalized when the speaker’s own side is presented as causally responsible, for instance through explicit references to “we”, “our government”, or the speaker’s own governing bloc. Attribution is coded as externalized when causal responsibility is assigned outside the speaker’s own side, including other domestic political actors and, more occasionally, institutions or extra-domestic constraints.

Examples. “Our government managed inflation effectively” is coded as credit and internalized. “The government failed to protect families from rising prices” is coded as blame and externalized for an opposition speaker. “Energy prices rose because of international shocks” is coded as blame and externalized. “We made difficult choices and stabilized employment” is coded as credit and internalized.

B Classifier Performance and Valence Threshold

This section documents the measurement performance of the computational pipeline. The stage-one classifier uses a thresholded decision rule for the valence class in order to reduce overprediction of valence while preserving overall classification quality.

Table 3 reports headline held-out performance for both stages of the pipeline. The tables that follow unpack the main three-class task by class, by country, by threshold choice, and by the resulting class distribution. The figures then visualize the held-out confusion patterns for stage one and for the two stage-two adapters.

Table 3: Held-out performance across the two-stage pipeline

Model	Accuracy	Macro F1	Key precision	Key recall
Stage 1, argmax rule	0.748	0.727	0.613	0.691
Stage 1, selected threshold rule	0.753	0.727	0.648	0.628
Stage 2, polarity	0.854	0.847	0.809	0.822
Stage 2, attribution	0.761	0.759	0.817	0.745

The key measurement pressure point is stage one. The selected 0.56 threshold reduces residual overprediction of valence while preserving overall classification quality. This matters substantively because the paper’s main dependent variable is the speech-level valence share.

Table 4: Stage-one held-out performance by class

Class	Precision	Recall	F1 score
Positional	0.672	0.792	0.727
Valence	0.613	0.691	0.650
Other	0.866	0.751	0.805

Table 5: Stage-one held-out performance by country

Country	Held-out N	Macro F1	Valence F1
AT	64	0.800	0.710
CZ	66	0.623	0.438
DE	52	0.757	0.741
DK	63	0.802	0.762
ES	51	0.740	0.750
HR	55	0.660	0.667
HU	82	0.675	0.579

Table 6: Threshold audit for the stage-one valence decision rule

Decision	Threshold	Accuracy	Macro F1	Valence precision	Valence recall	Predicted valence share
Baseline threshold rule	0.56	0.753	0.727	0.648	0.628	21.0%
Higher-recall alternative	0.38	0.739	0.721	0.581	0.723	27.0%

Table 7: Stage-one confusion focus: gold versus predicted class distribution

Class	Gold share	Predicted share	Precision	Recall	Predicted - gold (pp)
Positional	24.5%	28.9%	0.672	0.792	4.4
Valence	21.7%	24.5%	0.613	0.691	2.8
Other	53.8%	46.7%	0.866	0.751	-7.2

Table 8: Held-out performance for the stage-two adapters

Adapter	Accuracy	Macro precision	Macro recall	Macro F1	Key precision	Key recall
Polarity (credit vs blame)	0.854	0.846	0.848	0.847	0.809	0.822
Attribution (internalized vs externalized)	0.761	0.759	0.763	0.759	0.701	0.781

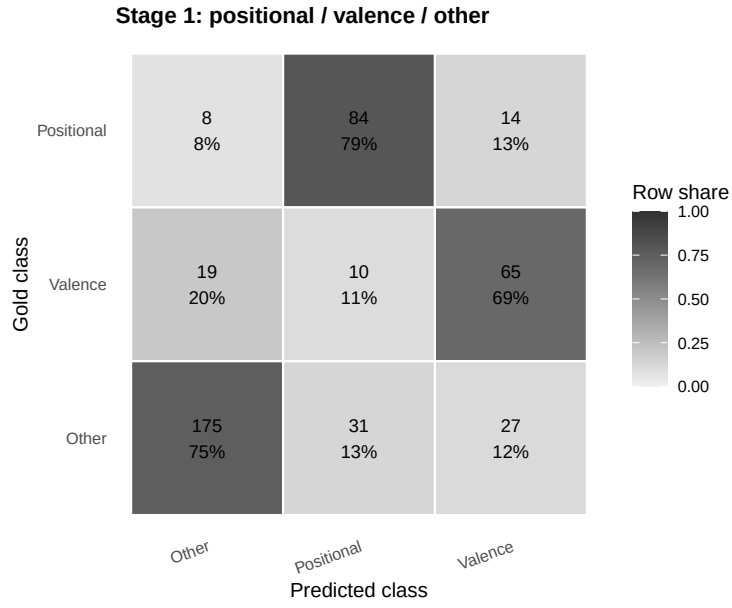


Figure 4: Stage-one confusion matrix on the held-out test set, shown as row-normalized shares.

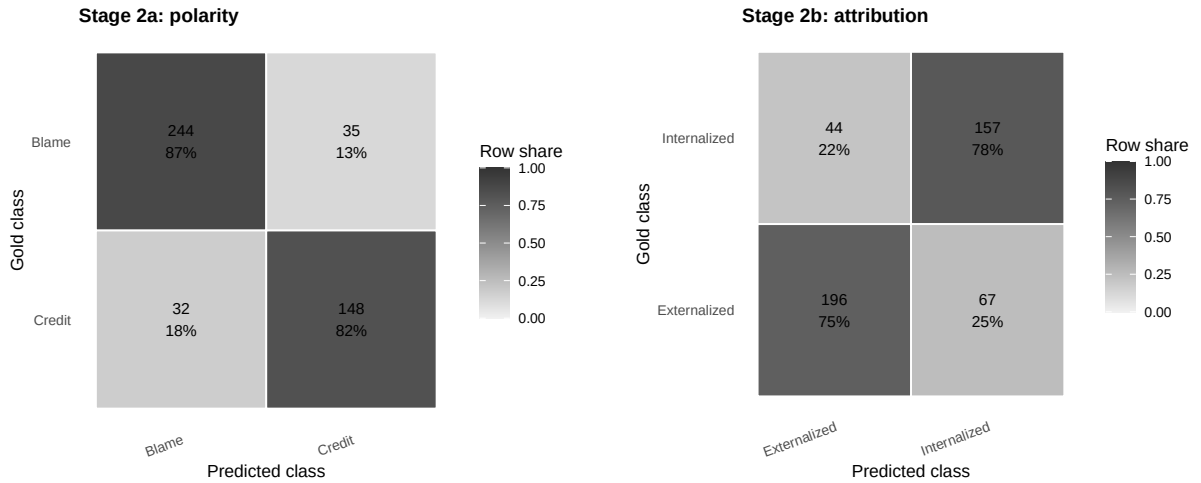


Figure 5: Held-out confusion matrices for the polarity and attribution adapters, shown as row-normalized shares.

C Sample Construction and Descriptive Evidence

This section documents the empirical base of the paper. It reports how the speech-level outcomes are distributed, how those outcomes differ descriptively by office status, and how the composition threshold structures the conditional sample.

Topical-gate validation

The topical gate is intentionally narrow. It is designed to capture explicit parliamentary debate over macroeconomic performance, not all speeches that bear indirectly on the economy. The appendix evidence therefore evaluates whether the rule captures that restricted domain cleanly and transparently. A deterministic dictionary rule is useful for that purpose because every inclusion decision can be traced to an observed lexical trigger in a known macroeconomic domain. By contrast, a topic model or LLM-based gate would move additional uncertainty into sample definition itself, making inclusion depend on prompts, thresholds, versions, or latent topics that are harder to audit.

Table 9 reports a manual quality-control check based on 560 labeled items. The QC sample was constructed as 40 rule-positive cases and 40 hard-negative cases per country, where the hard-negative block draws on long speeches and speeches with adjacent economic language but no rule match. The resulting perfect country-level scores therefore indicate that the rule behaves as expected on this targeted quality-control sample; they should not be read as the equivalent of a difficult held-out topical-classification benchmark. Table 10 compares the same rule-based filter with a Qwen-based topical classifier on overlapping country files. In this comparison, the Qwen classifications are materially narrower, which shows that model-based filtering is sensitive to implementation choices and should not be treated as a substitute for the rule-based measure.

Table 9: Manual quality-control check for the macroeconomic topical gate

Country	Precision	Recall	F1 score	QC sample N
AT	1.000	1.000	1.000	80
CZ	1.000	1.000	1.000	80
DE	1.000	1.000	1.000	80
DK	1.000	1.000	1.000	80
ES	1.000	1.000	1.000	80
HR	1.000	1.000	1.000	80
HU	1.000	1.000	1.000	80

Table 10: Rule-based topical gate compared with the LLM-assisted screen on overlapping country files

Country	Overlap	Agreement	Rule yes / LLM no	Rule no / LLM yes	Rule rate	LLM rate
AT	82,724	87.0%	13.0%	0.0%	13.0%	0.0%
CZ	19,587	87.6%	11.2%	1.2%	13.5%	3.4%
DE	47,284	85.8%	13.6%	0.6%	14.8%	1.8%
DK	40,006	83.1%	14.6%	2.3%	16.5%	4.1%

Multilingual keyword lists for the macro-topic gate

The deterministic gate is implemented through language-specific dictionaries for the three macroeconomic domains used in the paper: GDP/growth, inflation/prices, and unemployment/labour-market conditions. Table-free reporting is preferable here because the lists are short, exact, and meant to be auditable. The terms below are taken directly from the rule-based filtering script used in Step 1 of the pipeline. The purpose of the lists is to capture direct mentions of the three macroeconomic concepts and their close paraphrases across languages, not to classify broad policy agendas.

GDP / growth. *English:* gdp, gross domestic product, economic growth, growth rate, recession, economic contraction, real gdp.

Spanish: pib, producto interior bruto, producto interno bruto, crecimiento economico, tasa de crecimiento, recesion.

German: bip, bruttoinlandsprodukt, wirtschaftswachstum, wachstumsrate, rezession.

Danish: bnp, bruttonationalprodukt, okonomisk vaekst, vaekstrate, recession.

Croatian: bdp, bruto domaci proizvod, gospodarski rast, stopa rasta, recesija.

Hungarian: gdp, brutto hazai termek, gazdasagi novekedes, novekedesi rata, recesszio.

Czech: hdp, hruby domaci produkt, ekonomicky rust, tempo rustu, recese.

Inflation / prices. *English:* inflation, consumer price, cpi, cost of living, rising prices.

Spanish: inflacion, indice de precios, ipc, coste de la vida, subida de precios.

German: inflation, inflationsrate, teuerung, verbraucherpreisindex, lebenshaltungskosten.

Danish: inflation, forbrugerpriser, forbrugerprisindex, leveomkostninger, prisstigninger.

Croatian: inflacija, potrosacke cijene, indeks potrosackih cijena, troskovi zivota, rast cijena.

Hungarian: inflacio, fogyasztói arak, fogyasztói arindex, megelhetesi koltsegek, aremelkedes.

Czech: inflace, spotřebitelské ceny, index spotřebitelských cen, životní náklady, růst cen.

Unemployment / labour market. *English:* unemployment, joblessness, labour market, labor market, employment rate, unemployment rate.

Spanish: desempleo, paro, mercado laboral, tasa de paro, tasa de desempleo.

German: arbeitslosigkeit, arbeitslose, arbeitsmarkt, arbeitslosenquote, beschäftigungsquote.

Danish: arbejdsloshed, ledighed, arbejdsmarked, arbejdsloshedsrate, beskæftigelsesgrad.
Croatian: nezaposlenost, nezaposleni, trziste rada, stopa nezaposlenosti, stopa zaposlenosti.
Hungarian: munkanelkuliseg, munkaero piac, munkanelkulisegi rata, foglalkoztatottsagi rata.
Czech: nezamestnanost, trh prace, mira nezamestnanosti, mira zamestnanosti.

Construction and variation of the clarity index

The baseline moderator is designed to capture government clarity rather than the broader institutional environment. Table 11 reports the three ingredients of the index, their theoretical direction, their pooled distributions, and their empirical association with the final measure. The index is built as the equally weighted average of pooled-sample z -scores for fewer governing parties, the seat share of the prime minister’s party, and coalition seat concentration. This pooled standardization is used so that the three components enter on a common metric before aggregation and so that one unit of the final index corresponds to one standard deviation in the speech-level sample used for estimation.

Table 12 then shows that the three components point in the same empirical direction rather than canceling one another out. Their correlations with the final index range from 0.758 to 0.834. Figure 6 shows the pooled distribution of the index, while Figure 7 shows the country-quarter trajectories. The trajectories make clear that the index is not driven only by cross-country differences. It also moves within countries over time, especially around cabinet turnover, coalition entry and exit, and shifts in the internal balance of governing seats.

Table 11: Construction of the clarity-of-responsibility index

Component	Raw variable	Direction in index	Pooled mean	Pooled SD	Correlation with final index	Within-country SD (raw)
Fewer governing parties	n_gov_parties	Higher clarity when fewer parties share office	2.042	0.959	0.791	0.681
Prime minister party seat share	pm_party_seat_share	Higher clarity when the PM party controls more seats	0.404	0.117	0.758	0.072
Coalition seat concentration (HHI)	gov_hhi	Higher clarity when governing seats are concentrated	0.737	0.205	0.834	0.120

Table 12: Empirical alignment of the clarity components

Component	Fewer governing parties	PM party seat share	Coalition concentration	Final clarity index
Fewer governing parties	1.000	0.350	0.531	0.791
PM party seat share	0.350	1.000	0.455	0.758
Coalition concentration	0.531	0.455	1.000	0.834
Final clarity index	0.791	0.758	0.834	1.000

Country coverage

Table 13 reports the size of the speech-level panel in each country together with baseline averages for valence, clarity, and the macroeconomic controls used in the main specification.

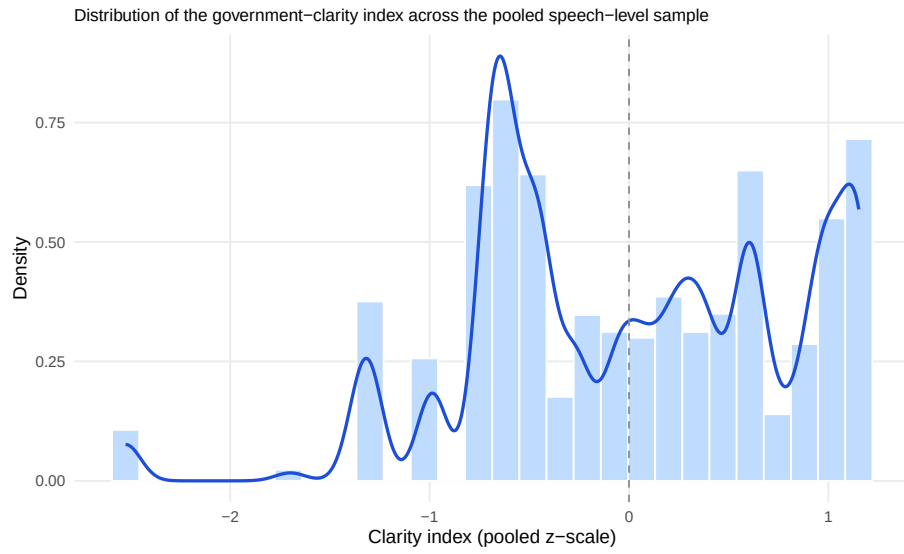


Figure 6: Distribution of the clarity-of-responsibility index in the pooled speech-level sample.

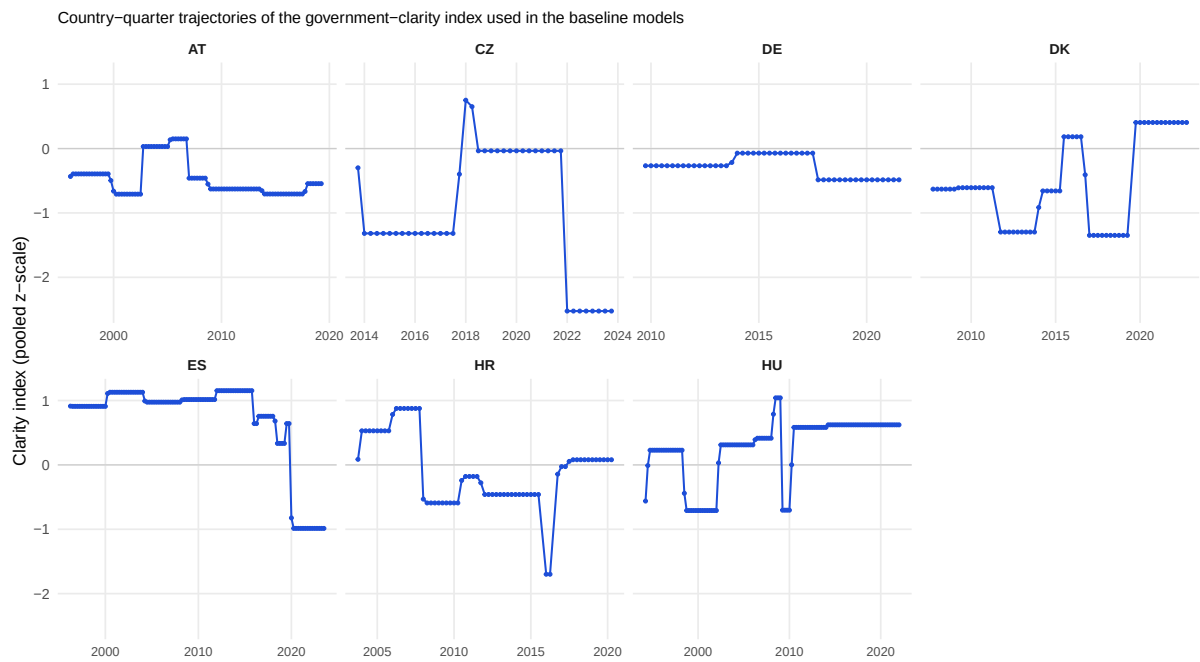


Figure 7: Country-quarter trajectories of the clarity-of-responsibility index.

Table 13: Country coverage and baseline speech characteristics

country	Speeches in panel	Gov share	Mean valence share	Mean clarity	Mean lagged GDP growth (QoQ)	Mean lagged inflation (QoQ)	Mean lagged unemp. change (QoQ)
AT	10,719	49.3%	0.255	-0.468	0.43	0.36	0.13
CZ	2,708	31.1%	0.215	-1.360	0.54	1.46	0.08
DE	7,134	58.7%	0.227	-0.284	0.46	0.32	-0.12
DK	9,657	36.4%	0.161	-0.592	0.25	0.44	0.14
ES	19,063	13.6%	0.296	0.737	0.43	0.39	0.06
HR	8,230	24.1%	0.264	-0.190	0.23	0.39	-0.15
HU	16,016	25.1%	0.201	0.247	0.73	1.25	-0.01

Speech-level descriptives

Table 14 reports the overall distribution of the speech-level outcomes. Table 15 then shows the raw government-opposition contrast before fixed effects are introduced.

Table 14: Speech-level outcome distribution

Statistic	Value
Mean positional share	0.309
Mean valence share	0.241
Mean other share	0.450
Mean valence share among substantive content	0.434
Mean credit share within valence	0.209
Mean internalized share within valence	0.292
Speeches in valence-share models	75,528
Speeches in composition models	65,257

Table 15: Speech-level means by office status

Office	Valence share	Credit share	Internalized share	N speeches
Government	0.169	0.336	0.401	22,427
Opposition	0.269	0.158	0.247	51,100

Composition-model selection

Table 16 shows how the composition-model threshold changes the estimation sample. The threshold excludes speeches with too little valence content to support stable within-valence shares.

Table 16: Composition-model selection

Group	N speeches	Mean valence share	Mean n valence qs
All macroeconomic speeches	75,528	0.241	8.664
Speeches with < 3 valence quasi-sentences	26,955	0.098	1.453
Speeches with $\geq$ 3 valence quasi-sentences	48,573	0.320	11.140

D Core Robustness for the Clarity-Centered Design

This section reports the checks most relevant to the paper’s baseline design, where clarity of responsibility is the central moderator and macroeconomic indicators are treated as controls.

Few-cluster inference

Because the baseline models cluster standard errors at the country level, the small number of country clusters deserves explicit attention. Table 17 therefore reports a country-level few-cluster check based on the distribution of country-specific interaction estimates, following the logic of Ibragimov and Muller (2016). The exercise is intentionally narrow: it focuses only on the main government-by-clarity interaction in the three baseline outcomes. The resulting pattern matches the main text. The valence-share interaction remains negative, and the composition interactions remain positive.

Table 17: Few-cluster check using country-specific interaction estimates

Outcome	Mean interaction	IM SE	IM p-value	Countries
Valence share	-0.033**	0.011	0.023	7
Credit share	0.055*	0.023	0.057	7
Internalized share	0.046***	0.011	0.006	7

Valence-share model on the composition-model sample

Table 18 compares the headline clarity interaction on the full valence-share sample and on the sample used for the composition outcomes.

Table 18: Valence-share clarity interaction on the full and composition-restricted samples

Sample	Observations	Gov x Clarity	SE
All valence-share speeches	67,930	-0.047**	0.013
Restricted to speeches with ≥ 3 valence quasi-sentences	42,926	-0.032**	0.013

Country-date fixed effects

Table 19 adds country-date fixed effects so that identification comes from government-opposition differences within the same country-day discussion windows.

Table 19: Country-date fixed-effects robustness

Term	Valence share	Credit share within valence	Internalized share within valence
In government	-0.145*** (0.017)	0.232*** (0.036)	0.203*** (0.027)
Gov x Clarity	-0.063** (0.023)	0.087 (0.049)	0.077* (0.038)

Leave-one-country-out re-estimation

Table 20 and Figure 8 show how the clarity interaction changes when each country is omitted in turn.

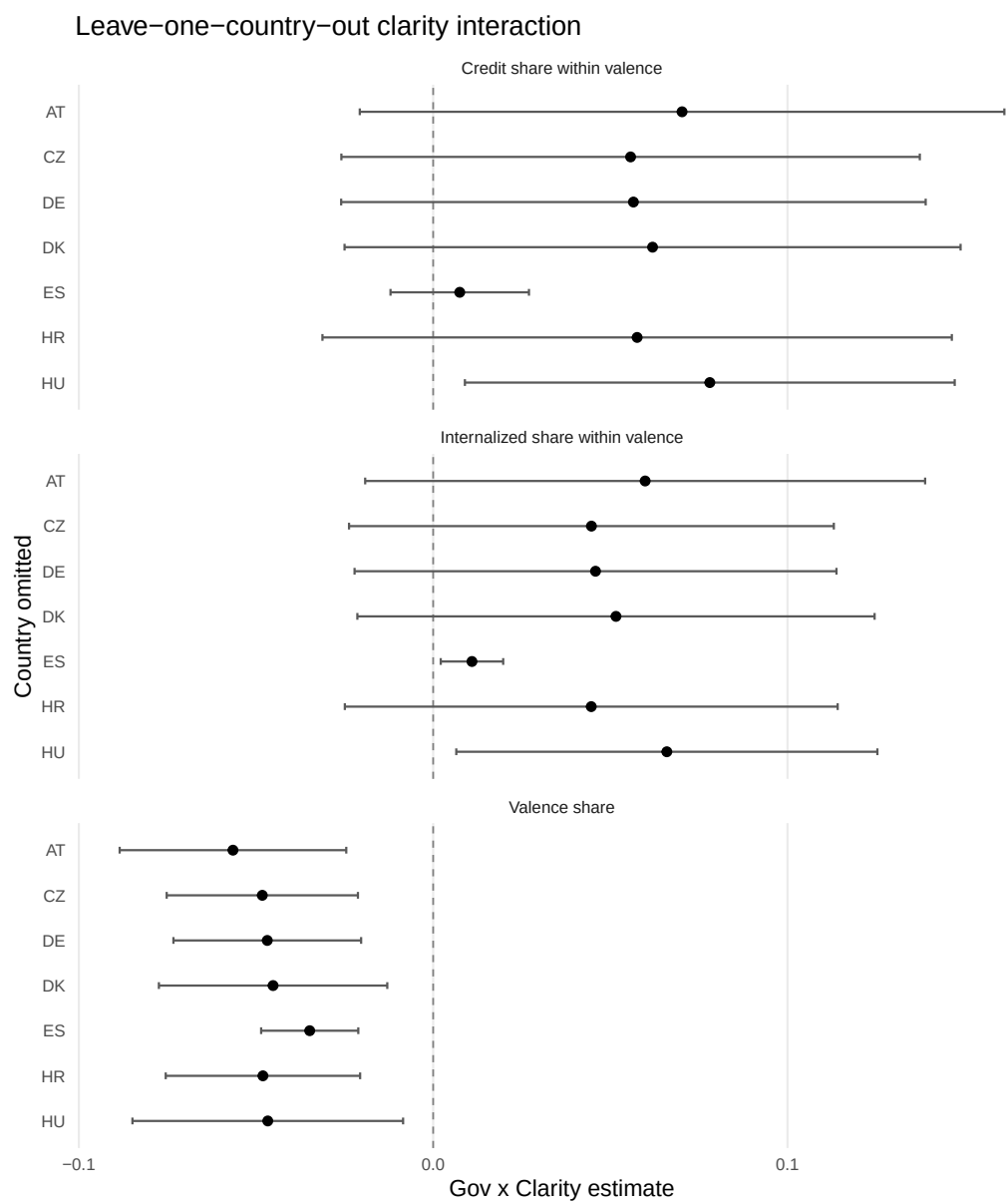


Figure 8: Leave-one-country-out estimates for the clarity interaction.

Table 20: Leave-one-country-out robustness for the clarity interaction

Omitted country	Outcome	Estimate	SE	N
AT	Valence share	-0.057**	0.016	57,603
AT	Credit share within valence	0.070	0.046	34,864
AT	Internalized share within valence	0.060	0.040	34,864
CZ	Valence share	-0.048**	0.014	66,325
CZ	Credit share within valence	0.056	0.042	42,021
CZ	Internalized share within valence	0.045	0.035	42,021
DE	Valence share	-0.047**	0.014	60,901
DE	Credit share within valence	0.057	0.042	37,241
DE	Internalized share within valence	0.046	0.035	37,241
DK	Valence share	-0.045**	0.016	58,308
DK	Credit share within valence	0.062	0.044	40,336
DK	Internalized share within valence	0.052	0.037	40,336
ES	Valence share	-0.035***	0.007	51,613
ES	Credit share within valence	0.008	0.010	31,382
ES	Internalized share within valence	0.011*	0.004	31,382
HR	Valence share	-0.048**	0.014	59,705
HR	Credit share within valence	0.058	0.045	37,609
HR	Internalized share within valence	0.045	0.035	37,609
HU	Valence share	-0.047*	0.019	53,125
HU	Credit share within valence	0.078*	0.035	34,103
HU	Internalized share within valence	0.066*	0.030	34,103

Within-party switcher check

Table 21 restricts the sample to parties that are observed both in government and in opposition. Figure 9 provides the corresponding descriptive comparison by country.

Table 21: Within-party switcher check

Sample	Term	Estimate	N
Switcher parties only	In government	-0.126*** (0.011)	37,786
Switcher parties only	Gov x Clarity	-0.043*** (0.010)	37,786

Intra-coalition variation among governing parties

The main design compares governments with oppositions, but governing parties are not homogeneous. This subsection restricts the sample to coalition cabinets and asks whether speeches by the prime minister’s party differ from speeches by junior coalition partners. The purpose is narrow: to verify that the main government-opposition result is not simply an artifact of junior coalition parties behaving differently inside government.

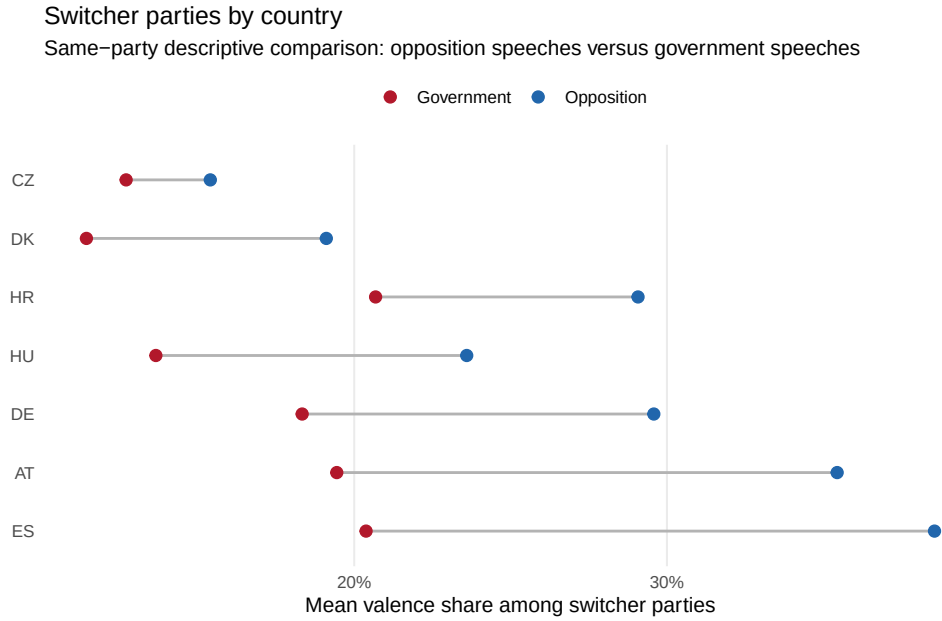


Figure 9: Descriptive switcher-party comparison by country: opposition versus government valence share.

Table 22: Descriptive comparison of PM-party and junior-coalition speeches

Coalition status	Speeches	Parties	Countries	Mean valence share	Mean credit share	Mean internalized share
Junior coalition	6,535	22	6	0.172	0.307	0.373
PM party	11,172	10	6	0.160	0.333	0.401

Table 23: PM-party versus junior-coalition contrast among governing speeches

Outcome	PM party	SE (PM party)	PM party x Clarity	SE (interaction)	N
Valence share	0.033*	0.013	0.065*	0.030	17,707
Credit share within valence	0.029	0.018	-0.026	0.025	14,380
Internalized share within valence	0.008	0.017	0.008	0.024	14,380

All-speech composition outcomes

Table 24 reports models in which credit and internalized attribution are expressed as shares of all quasi-sentences rather than as shares conditional on valence.

Table 24: All-speech composition outcomes

Term	Credit share of all speech	Internalized share of all speech
In government	0.029*** (0.007)	0.014* (0.006)
Gov x Clarity	0.011 (0.009)	0.008 (0.008)

Mirror and diagnostic outcomes, including positional share

Table 25 reports the mirror outcomes used to diagnose where the mass moves when overall valence changes. This includes the positional-share model discussed in the main text.

Table 25: Mirror and diagnostic outcomes

Term	Outcome	Estimate	N
In government	Positional share	0.059*** (0.006)	67,930
Gov x Clarity	Positional share	-0.001 (0.008)	67,930
In government	Other share	0.075*** (0.009)	67,930
Gov x Clarity	Other share	0.048*** (0.012)	67,930
In government	Valence share among substantive content	-0.181*** (0.013)	67,149
Gov x Clarity	Valence share among substantive content	-0.044** (0.012)	67,149
In government	Blame share within valence	-0.197*** (0.031)	42,926
Gov x Clarity	Blame share within valence	-0.055 (0.040)	42,926
In government	Blame minus credit within valence	-0.395*** (0.062)	42,926
Gov x Clarity	Blame minus credit within valence	-0.110 (0.080)	42,926

Alternative estimators

Table 26 compares the linear fixed-effects estimates with weighted quasi-binomial fixed-effects models.

Macroeconomic control sensitivity

Table 27 shows that the clarity interaction is stable across alternative ways of introducing macroeconomic controls.

Table 26: Alternative estimators for the main clarity-centered outcomes

Outcome	Model	Term	Estimate	N
Valence share	OLS FE	In government	-0.134*** (0.012)	67,930
Valence share	OLS FE	Gov x Clarity	-0.047** (0.013)	67,930
Valence share	Weighted quasi-binomial FE	In government	-0.815*** (0.038)	67,930
Valence share	Weighted quasi-binomial FE	Gov x Clarity	-0.209** (0.065)	67,930
Credit share within valence	OLS FE	In government	0.197*** (0.031)	42,926
Credit share within valence	OLS FE	Gov x Clarity	0.055 (0.040)	42,926
Credit share within valence	Weighted quasi-binomial FE	In government	1.310*** (0.116)	42,926
Credit share within valence	Weighted quasi-binomial FE	Gov x Clarity	0.333 (0.186)	42,926
Internalized share within valence	OLS FE	In government	0.172*** (0.026)	42,926
Internalized share within valence	OLS FE	Gov x Clarity	0.045 (0.033)	42,926
Internalized share within valence	Weighted quasi-binomial FE	In government	0.899*** (0.109)	42,926
Internalized share within valence	Weighted quasi-binomial FE	Gov x Clarity	0.244 (0.144)	42,926

Table 27: Sensitivity of the clarity interaction to alternative macro-control choices

Specification	Outcome	N	In government	Gov x Clarity	Within R2
No macro controls	Valence share	69,804	-0.132*** (0.013)	-0.045** (0.013)	0.059
No macro controls	Credit share within valence	44,176	0.193*** (0.033)	0.053 (0.040)	0.086
No macro controls	Internalized share within valence	44,176	0.168*** (0.027)	0.044 (0.034)	0.054
Raw YoY controls	Valence share	66,721	-0.133*** (0.013)	-0.048** (0.015)	0.060
Raw YoY controls	Credit share within valence	42,082	0.200*** (0.031)	0.058 (0.040)	0.091
Raw YoY controls	Internalized share within valence	42,082	0.173*** (0.026)	0.046 (0.034)	0.056
Raw QoQ controls	Valence share	67,930	-0.134*** (0.012)	-0.047** (0.013)	0.060
Raw QoQ controls	Credit share within valence	42,926	0.197*** (0.031)	0.055 (0.040)	0.088
Raw QoQ controls	Internalized share within valence	42,926	0.172*** (0.026)	0.045 (0.033)	0.055
Clean YoY index	Valence share	66,721	-0.133*** (0.012)	-0.046** (0.013)	0.059
Clean YoY index	Credit share within valence	42,082	0.199*** (0.031)	0.054 (0.038)	0.089
Clean YoY index	Internalized share within valence	42,082	0.173*** (0.025)	0.045 (0.032)	0.056
Clean QoQ index	Valence share	67,930	-0.133*** (0.012)	-0.046** (0.013)	0.060
Clean QoQ index	Credit share within valence	42,926	0.197*** (0.031)	0.054 (0.039)	0.088
Clean QoQ index	Internalized share within valence	42,926	0.172*** (0.026)	0.044 (0.033)	0.055

E Exploratory Heterogeneity

This section asks a narrower question than the main analysis: does the clarity pattern look similar across broad types of parties, or is it concentrated in a smaller subset of the party system? These splits are exploratory and descriptive. Their purpose is to show where the baseline pattern appears most strongly, not to replace the pooled comparative model.

Party-family heterogeneity

Table 28 reports the main valence-share model separately by broad party families.

Table 28: Exploratory heterogeneity by party family on the valence-share outcome

Family group	Term	Estimate	N
Conservative / Christian	In government	-0.090** (0.034)	18,440
Conservative / Christian	Gov x Clarity	-0.029 (0.021)	18,440
Social democratic	In government	-0.153*** (0.009)	15,859
Social democratic	Gov x Clarity	-0.040** (0.016)	15,859
Radical / Special issue	In government	-0.038*** (0.009)	8,695
Radical / Special issue	Gov x Clarity	0.007 (0.032)	8,695
Liberal	In government	-0.060*** (0.010)	6,432
Liberal	Gov x Clarity	0.014 (0.010)	6,432
Green	In government	-0.064* (0.029)	4,139

Populist-party split

The next table uses Zulianello’s PopulisTree taxonomy ([Zulianello, 2026](#)) to code populist status by speech year. It is treated only as a scope-condition check: the aim is to verify that the main clarity result is not driven by something special about populist parties.

Table 29: Zulianello split: populist versus non-populist speeches on the valence-share outcome

Group	Term	Estimate	N
Non-populist speeches	In government	-0.139*** (0.017)	60,260
Non-populist speeches	Gov x Clarity	-0.045** (0.015)	60,260
Populist speeches	In government	-0.096*** (0.009)	9,883
Populist speeches	Gov x Clarity	-0.032** (0.011)	9,883

Left-right split

Table 30 reports the same valence-share model split by broad left-right bloc.

Table 30: Left-right split on the valence-share outcome

Group	Term	Estimate	N
Left / center-left	In government	-0.139*** (0.011)	26,225
Left / center-left	Gov x Clarity	-0.030** (0.011)	26,225
Center-right / right	In government	-0.080*** (0.019)	32,163
Center-right / right	Gov x Clarity	-0.021 (0.012)	32,163